



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 5

1. Case Study Title: Implementation of JavaScript Array Operations Using Built-in Methods

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Array Operations</title>
  <style>
    body {
      background-color: lightblue;
    }
  </style>
</head>
<body>
<h2>JavaScript Array Operations</h2>
<label>Enter array elements:</label>
<input type="text" id="arrayInput">
<br><br>
<label>Value for Push:</label>
<input type="number" id="pushValue">
<br><br>
<label>Value for Unshift:</label>
<input type="number" id="unshiftValue">
<br><br>
<label>Splice Index:</label>
<input type="number" id="spliceIndex">
<label>Splice Value:</label>
<input type="number" id="spliceValue">
<br><br>
<label>Slice Start:</label>
```



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```
<input type="number" id="sliceStart">

<label>Slice End:</label>
<input type="number" id="sliceEnd">
<br><br>
<button onclick="arrayOperations()">Perform
Operations</button>
<p id="output"></p>
<script src="shreya_1196_case5.js"></script>
</body>
</html>
```

```
function arrayOperations() {
    let arr = document.getElementById("arrayInput").value
        .split(", ").map(Number);

    let pushValue =
Number(document.getElementById("pushValue").value);
    let unshiftValue =
Number(document.getElementById("unshiftValue").value);

    let spliceIndex =
Number(document.getElementById("spliceIndex").value);
    let spliceValue =
Number(document.getElementById("spliceValue").value);

    let sliceStart =
Number(document.getElementById("sliceStart").value);
    let sliceEnd =
Number(document.getElementById("sliceEnd").value);
    let output = "";
    output += "Original Array: " + arr + "<br><br>";
    arr.push(pushValue);
    output += "After push: " + arr + "<br>";
```



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```
// pop()
arr.pop();
output += "After pop: " + arr + "<br>";

// unshift()
arr.unshift(unshiftValue);
output += "After unshift: " + arr + "<br>";

// shift()
arr.shift();
output += "After shift: " + arr + "<br>";

// splice()
arr.splice(spliceIndex, 1, spliceValue);
output += "After splice: " + arr + "<br>";

// slice()
let newArray = arr.slice(sliceStart, sliceEnd);
output += "After slice: " + newArray + "<br><br>";

// Maximum and Minimum
let maximum = Math.max(...arr);
let minimum = Math.min(...arr);

output += "Maximum: " + maximum + "<br>";
output += "Minimum: " + minimum;

document.getElementById("output").innerHTML = output;
}
```



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3. Output:

← → 🔄 📄 File /Users/shreyamaske/javascript_exp/jsexp5/shreya_1196_case5.html

☐☐☐

JavaScript Array Operations

Enter array elements:

Value for Push:

Value for Unshift:

Splice Index: Splice Value:

Slice Start: Slice End:

Original Array: 2,34,56,77,81,45,11

After push: 2,34,56,77,81,45,11,90

After pop: 2,34,56,77,81,45,11

After unshift: 1,2,34,56,77,81,45,11

After shift: 2,34,56,77,81,45,11

After splice: 2,34,56,66,81,45,11

After slice: 66,81,45

Maximum: 81

Minimum: 2

4. Result/Conclusion:

Thus, various JavaScript array operations were successfully performed and the results were displayed using user input.